A Useful Mathematical Trick, Telescoping Series, and the Infinite Sum of the Reciprocals of the Triangular Numbers

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Last update: August 20, 2021

1 A Useful Mathematical Trick: $\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+1}$

This incredibly useful bit of mathematical magic works when the denominators of the two fractions are of the form n and $n+1$, their numerators are both 1, and $n > 0$. For example, something like

$$\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) = \frac{1}{12} \text{ and } \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}$$

So why does this work? Consider

$$\begin{aligned}
\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) &= \frac{1}{n \cdot (n+1)} && \text{\# multiply through, } 1 \cdot 1 = 1 \\
&= \frac{(n-n)+1}{n \cdot (n+1)} && \text{\# } (n-n) = 0 \text{ and } 0+1 = 1 \\
&= \frac{(n+1)-n}{n \cdot (n+1)} && \text{\# rearrange (+ is commutative and associative)} \\
&= \frac{(n+1)}{n \cdot (n+1)} - \frac{n}{n \cdot (n+1)} && \text{\# split into a difference} \\
&= \frac{1}{n} - \frac{n}{n \cdot (n+1)} && \text{\# cancel } (n+1) \text{ in the first term} \\
&= \frac{1}{n} - \frac{1}{n+1} && \text{\# cancel } n \text{ in the second term } \square
\end{aligned}$$

So we have

$$\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+1} \tag{1}$$

An example of the use of this trick in evaluating infinite series is Burkard Polster's (aka Mathologer) use of Equation 1 when describing how Euler's solution to the Basel problem works [1].

2 Telescoping Series

What is a telescoping series? Briefly, a telescoping series is a sum that is characterized by partial sums (called telescoping sums) contain pairs of consecutive terms which cancel each other, leaving only the first and final terms. This cancellation of adjacent terms is whimsically called "collapsing the telescope".

For example, consider the infinite series whose terms are described by Equation 1. This is perhaps the canonical example of a telescoping series, which can be seen as follows:

$$\begin{aligned}
\sum_{n=1}^{\infty}\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) &= \sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right) && \text{\# Equation 1} \\
&= \lim_{N\to\infty}\sum_{n=1}^{N}\left(\frac{1}{n}-\frac{1}{n+1}\right) && \text{\# we are interested in the limit} \\
&= \lim_{N\to\infty}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{N}-\frac{1}{N+1}\right)\right] \\
&= \lim_{N\to\infty}\left[1+\left(-\frac{1}{2}+\frac{1}{2}\right)+\left(-\frac{1}{3}+\frac{1}{3}\right)+\cdots+\left(-\frac{1}{N}+\frac{1}{N}\right)-\frac{1}{N+1}\right] \\
&= \lim_{N\to\infty}\left[1-\frac{1}{N+1}\right] && \text{\# collapse the telescope} \\
&= 1
\end{aligned}$$

A bit more formally, a telescoping series is the sum of terms of the form $(a_j - a_{j+1})$. As a result of the structure of its terms a telescoping series has the following property:

$$\begin{aligned}
\sum_{j=1}^{n}(a_j - a_{j+1}) &= (a_1-a_2)+(a_2-a_3)+(a_3-a_4)+\cdots+(a_n-a_{n+1}) && \text{\# expand terms} \\
&= a_1+(a_2-a_2)+(a_3-a_3)+\cdots+(a_n-a_n)-a_{n+1} && \text{\# group like terms} \\
&= a_1-a_{n+1} && \text{\# simplify}
\end{aligned}$$

3 Triangular Numbers

The example I want to explore here is the infinite sum of the reciprocals of the triangular numbers [4]. So first, what is a triangular number? The first six triangular numbers are depicted as triangles in Figure 1. We can see from the figure that a triangular number T_n is the sum of the integers from 1 to n.

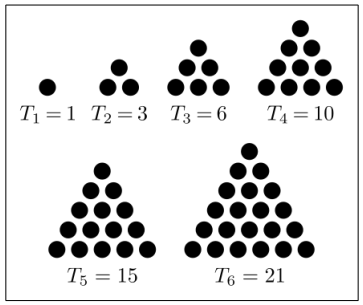

Figure 1: The First Six Triangular Numbers [4]

Since T_n is just the sum of the integers from 1 to n we can easily calculate the first few values of T_n, which are 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105 and so on (note that some authors include 0 in the triangular number sequence).

Ok, but what is the general form of T_n? Since T_n is the sum of the first n integers we know that[1]

$$T_n = \sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} = \binom{n+1}{2} \tag{2}$$

There is an interesting factoid involving the triangular numbers and Pascal's Triangle [3]. If you start with the number 1 in row 2 of Pascal's Triangle the corresponding diagonal contains the triangular numbers. This is shown in blue in Figure 2.

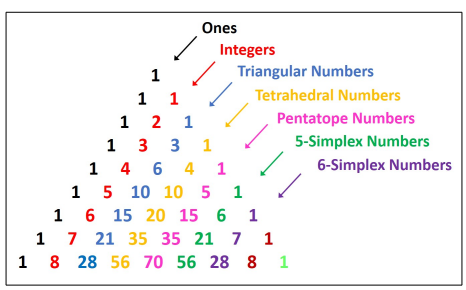

Figure 2: Pascal's Triangle

[1]The great German mathematician Carl Friedrich Gauss (1777-1855) is said to have discovered the consecutive integer formula while still at primary school [2].

4 The Sum of the Reciprocals of the Triangular Numbers

All of this is good, but back to the question: what is the sum of the reciprocals of the triangular numbers? Well, first, the sum looks like

$$\begin{aligned}
\sum_{n=1}^{N} \frac{1}{T_n} &= \sum_{n=1}^{N} \frac{1}{\frac{n(n+1)}{2}} && \text{\# definition of } T_n \text{ (Equation 2)} \\
&= \sum_{n=1}^{N} \frac{2}{n(n+1)} && \text{\# simplify} \\
&= 2\sum_{n=1}^{N} \left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) && \text{\# pull 2 out of the sum and group terms} \\
&= 2\sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) && \text{\# Equation 1}
\end{aligned}$$

We saw in Section 2 that $\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$ and we know from the above that the infinite sum of the reciprocals of the triangular numbers is twice this limit. So we know that

$$\sum_{n=1}^{\infty} \frac{1}{T_n} = 2\left[\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right)\right] = 2 \cdot 1 = 2$$

5 Conclusions

Equation 1 turns out to be very useful, especially when evaluating infinite telescoping series. We also saw in Section 4 that interestingly, the infinite sum of the reciprocals of the triangular numbers equals two, that is

$$\sum_{n=1}^{\infty} \frac{1}{T_n} = 2$$

As we saw in Section 2, the evaluation of this sum makes heavy use of a telescoping series described by Equation 1.

References

[1] Burkard Polster (aka Mathologer). Euler's real identity NOT e to the i pi = -1. https://www.youtube.com/watch?v=yPl64xi_ZZA, Aug 2017. [Online; accessed 17-August-2021].

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[4] Wikipedia contributors. Triangular Number — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Triangular_number&oldid=1034664425, 2021. [Online; accessed 17-August-2021].